

Multiple Choice (10 marks)

1. Which of the following sorting algorithms has average-case and worst-case running times of $O(n \log n)$?
 - (a) Bubble sort
 - (b) Insertion sort
 - (c) Merge sort
 - (d) Quicksort
2. Of the following, which gives the best upper bound for the value of $f(N)$ where f is a solution to the recurrence $f(2N + 1) = f(2N) = f(N) + \log N$ for $N \geq 1$, with $f(1) = 0$?
 - (a) $O(\log N)$
 - (b) $O(N \log N)$
 - (c) $O(\log N) + O(1)$
 - (d) $O((\log N)^2)$
3. Let $x = 8$. What is $x \gg 1$ in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8
4. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address
5. Of the following page-replacement policies, which is guaranteed to incur the minimum number of page faults?
 - (a) Replace the page whose next reference will be the longest time in the future.
 - (b) Replace the page whose next reference will be the shortest time in the future.
 - (c) Replace the page whose most recent reference was the shortest time in the past.
 - (d) Replace the page whose most recent reference was the longest time in the past.

True / False (10 marks)

Answer True or False

6. True or False: $O(1)$ is the smallest asymptotic notation, indicating constant time complexity regardless of input size.
7. True or False: Even after extensive testing, a program may still contain undetected bugs.
8. True or False: The condition $a < c$ is sufficient to ensure that the expression $a < c \parallel a < b \ \&\& \ ! (a == c)$ evaluates to true.
9. True or False: In Python 3, converting a list with duplicates to a set results in a collection of unique elements, such as $\{1, 2, 3, 4\}$.
10. True or False: The decimal number 0.5 has an exact representation in binary notation as 0.1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the number which occurs for odd number of times in the given array.
12. Write a function to sort each sublist of strings in a given list of lists using lambda function.

End of Paper